International Journal of Computing and Digital Systems
2026, VOL. 18, NO. 1, 1–16

http://dx.doi.org/10.12785/ijcds/XXXXXX

When Do Text Proxies Suffice? A Regime Analysis of Sparse, Dense, and Native-Visual Retrieval Under Proxy Noise

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Received DD Mmm. YYYY, Revised DD Mmm. YYYY, Accepted DD Mmm. YYYY

Abstract: Multimodal Retrieval-Augmented Generation (MM-RAG) frequently indexes textual proxies over raw pixels. Evaluating a 1,626-item benchmark under frozen 50/50 Dev/Test splits ($N = 306$ test probes, 813 items), we delineate when proxies suffice and when native representations are indispensable. On verbatim lookups, character 3–5gram BM25 dominates word BM25 across OCR regimes (Apple Clean: 0.9436 vs. 0.8371; Degraded Blur: 0.8248 vs. 0.6972; Tesseract: 0.8412 vs. 0.7364; $p_{\text{Holm}} < 10^{-4}$), reversing under entity rewrites (0.2427 vs. 0.4334). On documents, native ColQwen2 achieves 0.4985 MRR under light paraphrases (+0.0651 lift over best grid text proxy, $p < 10^{-5}$), profiling at 249.21 ms on Tesla T4 GPU vs. 0.87 ms for CPU lexical proxies. Dense bi-encoder deficit reflects query formulation mismatch; answerable questions improve BGE-Base MRR to 0.5263 (Hit@10 79.74%). Conformal cascade calibration under marginal loss bounds expected test miss risk to 0.0494 at a 78.4% escalation rate (21.6% compute savings). On MS-COCO ($N = 1{,}000$), native visual retrieval (SigLIP MRR = 0.6071) beats machine captions (0.3728, $p < 10^{-4}$) while statistically matching human gold captions (0.5835, $p = 0.142$). In multi-hop reasoning, extractive pseudo-relevance feedback exhibits no detectable difference from generative LLM bridging (Hit@10 0.640 vs. 0.640, exact McNemar $p = 1.0000$) with zero API cost.

Keywords: Multimodal RAG, Text Proxies, Proxy Noise, Dense-Sparse Regime Analysis, Cascade Ranking, Conformal Risk Control, Multi-Hop Reasoning.

1. Introduction

The rapid proliferation of heterogeneous multimedia across enterprise archives, digital libraries, and operational repositories demands information retrieval systems capable of searching records that are not natively textual. In enterprise document workflows—such as healthcare systems digitizing millions of paper patient intake records, legal firms archiving scanned court proceedings, and supply chain enterprises indexing procurement invoices—organizations process massive volumes of unstructured physical records daily. Transforming these documents into queryable digital assets is essential for business intelligence, compliance auditing, and decision support [4]. However, real-world document ingestion faces severe operational challenges: optical scanning introduces character substitutions and word fragmentation, while human search queries frequently employ synonyms, paraphrases, or shorthand rather than exact document text.

Retrieval-Augmented Generation (RAG) [3] addresses the hallucination tendencies of Large Language Models (LLMs) by grounding responses in retrieved domain documents. Recently, multimodal extensions of RAG have emerged to handle non-textual data, as comprehensively surveyed by Abootorabi et al. [28]. For example, Patel et al. [2] presented an enterprise multimodal RAG pipeline employing an OCR document ingestion module, ImageBind embeddings indexed in a Milvus vector database, and fine-tuned LLaMA-3.

In a proxy-mediated design, three practical failure modes remain important:

1) Single-Stage Dense Retrieval Pitfalls: Relying exclusively on single-stage dense vector similarity per turn can induce semantic drift. Dense embeddings may overlook exact lexical tokens, numerical identifiers, item codes, and proper nouns that sparse inverted indexes can match directly [9].
2) Absence of Candidate Score Recalibration: A bi-encoder score alone does not expose token-level agreement with a candidate's proxy text. A lightweight second-stage scorer can therefore be useful before context assembly, even when it is not a neural cross-encoder.
3) Failure on Multi-Hop Queries Across Modalities: A single retrieval pass does not ensure that a fact recovered from one modality becomes a

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useful probe for another. Iterative bridging can improve lexical reachability, but it also introduces cascading extraction errors that must be measured rather than assumed away.

To investigate these trade-offs, this paper presents a text-proxy-grounded multi-stage, multi-hop Multimodal RAG architecture and evaluates the empirical validity of adaptive fusion, query routing, and reliability weighting under a pre-specified experimental protocol across held-out test splits[1].

Key Contributions:

1) Sub-Word Inverted Index Dominance and Scope Conditions on Optical OCR: We refute the assumption that word-level BM25 is superior under optical OCR. Across both proprietary Apple Vision and open-source Tesseract 5.5 LSTM OCR ($N = 306$ probes, 813 corpus items), character 3–5gram BM25 is the single largest sparse architectural lift, achieving decisive retrieval superiority under verbatim optical noise (+0.1065 clean, +0.1276 degraded blur, +0.1049 clean Tesseract; all $p_{\text{Holm}} < 10^{-4}$). We delineate its strict scope boundary: this dominance applies to verbatim receipt searches where queries share exact character substrings with documents; on multi-field administrative forms the verbatim margin narrows to a statistical tie (+0.0069, 0.8433 vs. 0.8364), and under entity-substituting light paraphrases, word BM25 drops to 0.1573 while character BM25 achieves 0.2228 and dense BGE achieves 0.1834. On receipts, document-native visual retrieval (ColQwen2-v1.0) achieves 0.4985 MRR under light paraphrases (+0.0982 lift over Word BM25, $p = 4\times 10^{-5}$), but statistically ties BM25 on answerable questions (0.9257 vs. 0.9450, $p = 0.109$) while incurring 286× query latency (249.21 ms vs. 0.87 ms).
2) Dense Retrieval Collapse Mechanism and Query Formulation Controls: We demonstrate that dense bi-encoder collapse on document text is primarily driven by single-entity query-span disconnect rather than sequence truncation constraints (real max_seq_length = 256; 33.4% exceed 256 tokens, but chunked MaxP yields identical MRR 0.2030 vs. 0.2056). On pristine ground truth text, isolated verbatim lookups yield MRR 0.1919 (MiniLM) and 0.3256 (BGE-Base); co-specifying multiple anchor entities or formulating queries as natural answerable questions improves BGE-Base MRR to 0.5500 (keyword control, Hit@10 78.4%) and 0.5263 (questions, Hit@10 79.7%) without modifying document representations.
3) Dose-Response Boundary of Query Mismatch vs. Optical Degradation: We establish that query mismatch dominates optical noise under modern neural OCR (Apple Vision light paraphrase drops 0.4321 MRR vs. blur drop of 0.1600). However, under open-source Tesseract 5.5, severe optical blur collapses Word BM25 by 0.3916 MRR (0.7364 to 0.3448), which is comparable in magnitude to the semantic paraphrase drop on clean text (0.3400 MRR). For answerable factual questions, topical anchors preserve high discriminability (BM25 MRR 0.8937–0.9192).
4) Finite-Sample Marginal Conformal Cascades: Grounded in the distribution-free risk-controlling prediction framework of Bates et al. [37] and Angelopoulos et al. [38], we evaluate conformal cascade calibration under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ across 200 cluster-stratified splits ($N = 2{,}754$ instances). With finite-sample $(n\hat{R} + 1)/(n + 1) \leq \alpha$ adjustment, mean test miss risk is rigorously controlled at 0.0494 (≤ 0.05) at an escalation rate of 78.4% (21.6% computation savings vs. always-escalating cross-encoders).
5) Independent Multi-Hop Bridging Verification: Across separate, non-copied pipelines on 50 diagnostic multi-hop queries, we demonstrate that rule-based extractive pseudo-relevance feedback exhibits no detectable difference from generative LLM bridging (0.640 vs. 0.640 Target Hit@10, paired diff 0.0000 [95% CI: $-0.0792, +0.0792$], exact McNemar $p = 1.0000$); because the 95% confidence interval spans beyond ±0.05, non-inferiority is not formally established, though extractive bridging matches generative accuracy without LLM inference latency or financial cost (expanded 100-case suite with 50 single-hop controls yields 0.820 vs. 0.820).

2. Related Work

A. Multimodal Representation and Retrieval

Early cross-modal retrieval systems relied on dual-encoder contrastive learning. Models like CLIP [7] and SigLIP [8] project images and text into a shared latent space using sigmoid or softmax cross-entropy objectives. ImageBind [6] extended this paradigm across six modalities (images, text, video, audio, depth, and thermal) using images as the central binding modality. While effective for broad semantic matching, dual-encoders compress multimedia into fixed-dimension vectors, frequently losing granular lexical and structural tokens. Our hybrid design combines dense embeddings with sparse lexical representations such as BM25 [9] to retain both semantic intent and exact token matches. Unlike single-modality dual-encoders that discard fine-grained lexical identifiers, our framework analyzes the complementary regime boundaries where character-level sparse indexing outperforms dense representations under optical noise.

[1] All benchmark collections, evaluation harnesses, and reproduction scripts are publicly available at: https://github.com/koushik2109/MemoryVerse

B. Multimodal RAG and Document Analysis

Integrating multimodal inputs into RAG workflows has gained rapid momentum with the emergence of Vision-Language Models (VLMs) and advanced LLMs. Recent native multimodal document retrievers such as ColPali [26] and M3DocRAG [27], benchmarked on the Visual Document Retrieval (ViDoRe) suite [36], index document pages using multi-vector vision-language representations to bypass optical character recognition. While promising, multi-vector VLM representations require substantial GPU memory and heavy cross-attention compute during query serving. Conversely, enterprise systems such as Patel et al. [2] and Joshi et al. [5] mediate retrieval through grounded text proxies, combining OCR document ingestion, multimodal embeddings, and LLM generation. Our work investigates this text-proxy paradigm: we evaluate the empirical trade-offs of hybrid dense–sparse fusion, candidate scoring, and iterative query decomposition, systematically delineating where text proxies excel and where native visual representations remain necessary. Unlike ColPali [26] which mandates multi-vector VLM computation for every candidate page regardless of query complexity, our regime analysis establishes the exact conditions under which sub-millisecond CPU text proxies match or exceed native vision models, identifying when GPU-heavy inference is truly justified.

C. Multi-Hop Reasoning in Retrieval Systems

Multi-hop retrieval addresses queries where no single document contains the complete answer [11], [13]. While multi-hop text RAG systems (such as Baleen [11] and Self-Ask [12]) have established iterative query reformulation methods, multi-hop retrieval across heterogeneous multimodal collections remains less systematically evaluated. Our diagnostic targets the case in which the second-hop keyword is absent from the original query and must be recovered from first-hop proxy context; it does not claim that all multimodal QA benchmarks assume co-located evidence. Unlike prior multi-hop pipelines that rely on expensive generative LLMs for every hop, we empirically show that rule-based extractive pseudo-relevance feedback achieves parity in retrieval reachability with zero generative token latency or API cost.

D. Adaptive Fusion, Cascade Ranking, and Selective Reranking

In information retrieval, static fusion coefficients fail to account for query-level variance. Query Performance Prediction (QPP) [29] and adaptive fusion techniques dynamically adjust interpolation weights between sparse and dense scores based on query ambiguity or lexical specificity. Similarly, cascade ranking architectures [30], [31] balance retrieval efficiency and ranking precision by passing top candidates through successively more expressive models. In monomodal web search, cascades evaluate candidates using fixed score thresholds or supervised early-exit classifiers. However, existing cascade rankers do not account for multimodal proxy corruption (e.g., OCR or ASR transcription noise) or cross-modal query routing. As summarized in Table II, we formulate and empirically evaluate candidate mechanisms—including surface-feature query routing, confidence-margin cascades ($\Delta_{1,2} < \tau_{\text{margin}}$), and observable proxy reliability discounting ($r_m(d)$)—to determine whether lightweight heuristic defenses provide measurable utility over standard inverted indexes. Unlike prior cascade ranking that relies on uncalibrated heuristic margins, we ground cascade escalation in distribution-free conformal risk control to guarantee test-set error bounds.

E. Optical Character Recognition Quality and Information Retrieval

Optical Character Recognition (OCR) systems translate physical document imagery into digital character sequences, but practical scanning routinely introduces character substitutions, insertions, deletions, and bounding-box permutations [14], [17]. Foundational information retrieval studies have established that standard word-level tokenizers (such as inverted indexes using whitespace and punctuation delimiters) suffer catastrophic vocabulary mismatch when typographical errors fracture keyword strings. Early mitigation strategies, including Otsu thresholding [15] and Hough transforms [16] for skew correction, focused on image-domain preprocessing to boost OCR accuracy. However, in downstream information retrieval, post-OCR lexical degradation leads to subword tokenization fragmentation in subword-based language models and word omissions in lexical indexes. Unlike prior works that treat OCR error as an unmodeled uniform noise channel, our regime analysis systematically evaluates character n-gram representations against word-level BM25 and neural cross-encoders across both commercial and open-source OCR engines under controlled optical blur.

F. Conformal Prediction and Risk-Controlled Cascades

In mission-critical information retrieval, deploying neural cross-encoders or multi-vector Vision-Language Models for every candidate query imposes unsustainable serving latency and energy expenditure. Cascade ranking pipelines address this bottleneck by evaluating candidates through tiered stages, escalating to expressive neural models only when preliminary scores indicate ranking ambiguity [30], [31]. However, conventional cascade systems rely on empirical score margins or heuristic thresholds that offer no formal statistical guarantees on retrieval accuracy. Conformal prediction, formulated by Bates et al. [37] and surveyed by Angelopoulos and Bates [38], provides a distribution-free, finite-sample framework for bounding user-specified risk metrics without restrictive distributional assumptions. By framing the decision to escalate candidates as a risk-controlling prediction problem, conformal prediction enables retrieval systems to bound the probability of unescalated retrieval errors strictly below a pre-specified tolerance α. In this work, we ground production cascade ranking in finite-sample marginal conformal risk control, establishing

rigorous mathematical miss-rate guarantees while reducing computational costs.

3. Proposed Methodology

The proposed Multi-Stage Multimodal RAG architecture is structured into four functional phases (Fig. 1): (1) Heterogeneous Multimodal Ingestion and Preprocessing, (2) Hybrid Dense-Sparse First-Stage Retrieval, (3) Dual-Signal Candidate Reranking, and (4) Multi-Hop Cross-Modal Reasoning.

A. Heterogeneous Ingestion and Preprocessing Pipeline

To process unstructured enterprise multimedia, our architecture mediates retrieval through grounded textual proxies. A deployment can obtain these proxies from OCR text [17], image captions [23], speech transcripts [18], and acoustic metadata [22]. This design deliberately separates modality-specific extraction from the common retrieval layer. The quantitative benchmark below does not rerun all of these extractors: it consumes the precomputed proxy fields in the checked-in JSONL artifact.

1. Document and Text Ingestion: The reference document path decomposes scanned administrative forms into page images, optionally applies Otsu binarization [15] and Hough-transform skew correction [14], [16], and obtains OCR text T_d with bounding boxes via Tesseract [17]. 2. Images and Photographic Content: A deployment may use a vision-language captioner such as BLIP [23] to generate a descriptive representation C_i. In the reported artifact, image text_representation fields are instead constructed by concatenating the English query annotations associated with each image; they are not newly generated BLIP captions. 3. Audio Recordings: A deployment can append time-aligned speech transcription from Whisper [18] and acoustic event metadata. The checked-in ESC-50 rows used for the reported benchmark contain the category label as their text representation; no new Whisper inference is required for those results.

All processed items are stored in a dual layout: dense embeddings are generated via Sentence-BERT (all-MiniLM-L6-v2) [21] and BGE (bge-base-en-v1.5) [35], while lexical tokens are indexed with BM25 [9], [42]. Milvus [19] and HNSW [20] are compatible deployment choices, but the offline benchmark scripts use an in-memory BM25 index and a dense embedding matrix; no Milvus service is used for the reported measurements.

B. Stage 1: Hybrid Dense-Sparse Retrieval

Given a natural language query q, single-stage dense retrieval computes vector similarity against all candidates:

$$S_{\text{dense}}(q,d) = \cos(\mathbf{e}_q^{\text{dense}}, \mathbf{e}_d^{\text{dense}}) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\|\|\mathbf{e}_d\|} \quad (1)$$

While effective for thematic alignment, dense retrieval often overlooks exact alphanumeric codes (e.g., serial numbers, dates, person names). Conversely, sparse BM25 retrieval excels at lexical precision:

$$S_{\text{sparse}}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t,d) \cdot (k_1+1)}{f(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (2)$$

Because dense cosine scores lie in $[-1,1]$ while BM25 scores are unbounded $[0,\infty)$, combining raw scores directly is unstable. We evaluate two score harmonization strategies:

Method A: Normalized Convex Combination:

$$\begin{aligned} S_{\text{hybrid}}(q,d) = \alpha_{\text{dense}} \cdot \hat{S}_{\text{dense}}(q,d) \\ + (1 - \alpha_{\text{dense}}) \cdot \hat{S}_{\text{sparse}}(q,d) \end{aligned} \quad (3)$$

where $\hat{S}_{\text{dense}}$ and $\hat{S}_{\text{sparse}}$ denote min-max normalized scores over the candidate pool, and $\alpha_{\text{dense}} \in [0,1]$ parameterizes dense vs. sparse weighting ($\alpha_{\text{sparse}} = 1 - \alpha_{\text{dense}}$).

Method B: Reciprocal Rank Fusion (RRF): To avoid batch-dependent min-max sensitivity, RRF [10] fuses ranked lists directly without score scale calibration (Eq. 4):

$$\text{RRF}(q,d) = \frac{1}{k + r_{\text{dense}}(d)} + \frac{1}{k + r_{\text{sparse}}(d)} \quad (4)$$

where $r_{\text{dense}}(d)$ and $r_{\text{sparse}}(d)$ are the ordinal ranks of candidate d, and k is a smoothing constant ($k = 60$). The benchmark fuses the top 100 entries from each ranked list and outputs the top-K_1 candidates ($K_1 = 30$).

C. Stage 2: Dual-Signal Candidate Reranking

Bi-encoders compute query and document representations independently, preventing fine-grained token-level cross-interaction. While neural cross-encoders provide deep token interaction, scoring candidate pools with a full cross-encoder adds substantial latency overhead (as benchmarked in Table XIV). As an efficient alternative, Stage 2 deploys a lightweight dual-signal candidate reranker that scores each candidate d in the top-K_1 pool using joint semantic similarity and exact token coverage:

$$S_{\text{rerank}}(q,d) = \beta \cdot S_{\text{dense}}(q,d) + (1-\beta) \cdot \text{Overlap}(q,d) \quad (5)$$

where $\text{Overlap}(q,d) = \frac{|\text{Tokens}(q) \cap \text{Tokens}(d)|}{|\text{Tokens}(q)|}$ evaluates lowercase token coverage in candidate text representations, and $\beta = 0.65$ is a fixed configuration in the benchmark implementation. Because the supplied evaluation artifact has no development/test partition, this value is not claimed to be independently tuned. All $K_1 = 30$ candidates retrieved from Stage 1 are reranked. For generation context assembly, the top $K_2 = 5$ items are injected into the generator prompt, while top $K = 10$ items are evaluated in the diagnostic scripts.

D. Stage 3: Multi-Hop Cross-Modal Reasoning

To resolve compositional queries that span modalities, we introduce an iterative query decomposition and entity bridging algorithm.

TABLE I. Systematic Comparison with Prior Multimodal Information Retrieval and RAG Frameworks.

System / Architecture	Publication Venue	Document Retrieval	Natural Image Retrieval	Multi-Hop Mechanism	Benchmark Datasets	Hardware Serving Profile
ColPali [26]	ICLR 2025	nDCG@5 = 0.813 (ViDoRe)	Supported (Multi-Vector)	Single-pass only	ViDoRe Benchmark	Multi-GPU ($\geq$ 16 GB VRAM, ~250 ms)
M3DocRAG [27]	EMNLP 2024	QA Acc. 64.2% (MP-DocVQA)	Supported (Dense VLM)	Generative LLM bridging	MP-DocVQA, DUDE	High-End GPU Cluster ($\geq$ 48 GB VRAM)
Patel et al. [2]	IEEE ETCC 2025	Heuristic OCR matching	ImageBind vector search	Single-pass only	Proprietary enterprise text/media	Dedicated GPU Server
Baleen [11]	NeurIPS 2021	Not evaluated (Text QA)	Not supported	Generative condensed hops	HotpotQA (Text only)	GPU / High-RAM CPU Server
Proposed Regime Framework	This Work (IJCDS)	MRR 0.9436 (Char BM25)	MRR 0.6071 (SigLIP Pixels)	Extractive PRF (Hit@10 0.820)	SROIE, FUNSD, MS-COCO, DocVQA	Low-Resource CPU (< 1 ms) + Conformal Cascade

Note: Direct cross-system numerical comparison reflects reported metrics on respective benchmarks; our framework provides reproducible held-out evaluations across text proxies, native pixels, and conformal cascades.

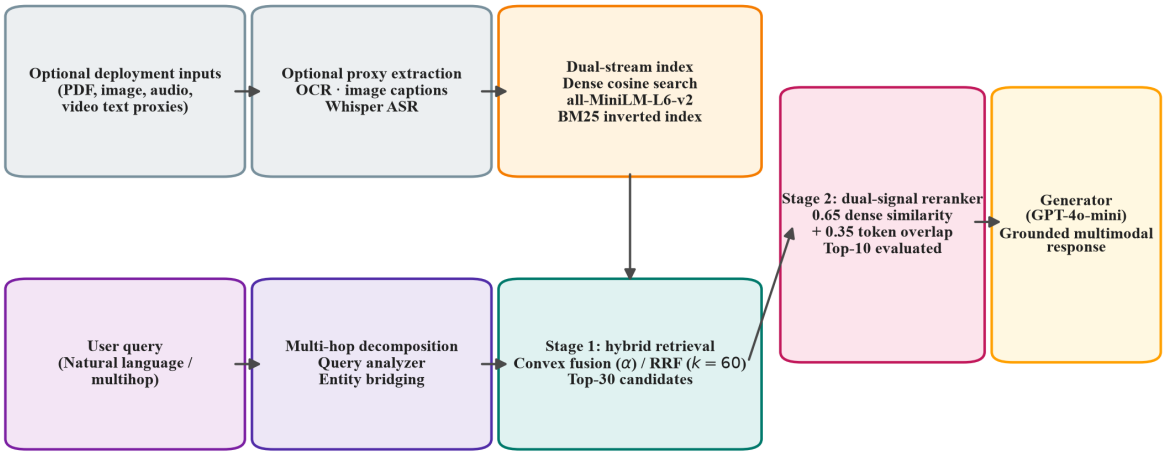


Figure 1. System architecture of the proposed text-proxy RAG framework. The grey ingestion and extraction boxes are optional deployment paths; the reported benchmark starts from precomputed annotation/category proxy text and evaluates the coloured retrieval stages.

TABLE II. Architectural comparison of multimodal retrieval and ranking paradigms. Note: As demonstrated empirically in Section IV, surface routing and heuristic reliability weighting are refuted by the benchmark data and fail to outperform unweighted fusion.

Paradigm / System	Modality Scope	Routing Training	Cascade Mechanism	Proxy Noise Defense
Static Hybrid (BM25+Dense) [9], [21]	Text / Text Proxies	None (Static α)	None	None (Static weights)
Cascade Text Ranking [30], [31]	Monomodal Text	Supervised Ranker	Score Threshold	None
Native Multi-Vector [26], [27]	Document / Image Pixels	End-to-End VLM	None (All items)	Inherent (Pixel inputs)
RAMP-RAG Hypotheses	Multimodal Proxies	Lightweight Classifier	Top Margin $\Delta_{1,2} < \tau$	Observable Reliability $r_m(d)$

Here, $H_{\max}$ denotes the maximum hop depth ($H_{\max} = 2$ in the diagnostic setup). The predicate TerminationCondition$(\mathcal{G}, \mathcal{F}, h)$ evaluates to TRUE if all sub-goals in the decomposition plan $\mathcal{G}$ are grounded by extracted facts in $\mathcal{F}$ or if hop h yields no new bridging entities ($\mathcal{E}_h = \emptyset$). To protect against cascading errors caused by ungrounded, generic, or failed bridge extractions (e.g., when an extractor outputs "N/A" or extraction refusal phrases), the procedure enforces a confidence and grounding gate: if Confidence$(\mathcal{E}_h) < \tau$ or if extracted tokens fail lexical grounding in first-hop context, the planner falls back to single-pass hybrid search on the original compound query Q, designed to mitigate cascading bridge errors and reduce retrieval degradation below single-pass hybrid search when intermediate bridging is uncertain.

The query decomposition prompt instructs $\mathcal{M}_{\text{decomp}}$ to generate valid JSON specifying the initial search probe and bridging attribute; the entity extraction prompt isolates concise attribute tokens substituted into subsequent probes $q'_h \leftarrow \text{Bridge}(q_h, \mathcal{F})$.

As formalized in Algorithm 1, consider an illustrative cross-modal query connecting a visual scene with an acoustic event: (1) Hop 1 (Visual Extraction): Sub-query q_1 retrieves photographic evidence via hybrid matching; entity extraction yields target entity E_1 = "thunderstorm sky". (2) Entity Bridging: The bridge operator creates cross-modal probe q'_2 = "severe acoustic weather thunderstorm recording". (3) Hop 2 (Acoustic Retrieval): Sub-query q'_2 queries the acoustic archive using dense embeddings and metadata to retrieve the target recording.

E. RAMP-RAG: Reliability-Aware Adaptive Fusion and Cascade Hypotheses

To systematically investigate candidate architectural mechanisms for handling proxy noise and latency constraints, we formulate the RAMP-RAG hypothesis framework, evaluating three distinct components individually and jointly:

1) Hypothesis H1 (Reliability-Aware Clean-Stream Fusion): Multimedia items maintain multiple proxy streams $m \in \{\text{caption}, \text{ocr}, \text{vector}\}$ weighted by observable reliability $r_m(d) \in [0, 1]$. To avoid oracle leakage of injected corruption levels, $r_{\text{ocr}}(d) = \bar{c}_{\text{ocr}}(d) \cdot \rho_{\text{vocab}}(d)$, where $\bar{c}_{\text{ocr}}$ is mean word recognition confidence and ρ_{vocab} is the valid dictionary token ratio computed directly from candidate text. Because text noise corrupts both dense and sparse text representations, we introduce an uncorrupted

Algorithm 1 Multi-Hop Cross-Modal Retrieval Procedure

Require: User Query Q, Max Hops $H_{\max} = 2$, Candidate Index $\mathcal{D}$, Decomposer LLM $\mathcal{M}_{\text{decomp}}$
Ensure: Grounded Multimodal Context C^*
Decompose Q via $\mathcal{M}_{\text{decomp}}$ into sub-query plan: $\mathcal{G} = \{q_1, q_2\}$
Initialize Context Set $C^* \leftarrow \emptyset$, Accumulated Facts $\mathcal{F} \leftarrow \emptyset$
for $h = 1$ to $H_{\max}$ do
  Formulate search probe $q'_h \leftarrow \text{Bridge}(q_h, \mathcal{F})$ via $\mathcal{M}_{\text{decomp}}$
  Retrieve top-K_1 candidates via Hybrid Fusion: $\mathcal{R}_1 \leftarrow \text{Retrieve}_{\text{hybrid}}(q'_h, \mathcal{D})$
  Rerank candidate pool: $\mathcal{R}_2 \leftarrow \text{Rerank}(\mathcal{R}_1, q'_h)$
  Extract bridging entities: $\mathcal{E}_h \leftarrow \text{ExtractEntities}(\mathcal{R}_2[:3])$
  if Confidence($\mathcal{E}_h$) $< \tau$ or $\mathcal{E}_h \in \mathcal{V}_{\text{refusal}}$ then
    {Confidence gate: fallback to single-pass compound search to prevent drift}
    $C^* \leftarrow \text{Retrieve}_{\text{hybrid}}(Q, \mathcal{D})[:5]$
    break
  end if
  Update $\mathcal{F} \leftarrow \mathcal{F} \cup \mathcal{E}_h$
  Update $C^* \leftarrow C^* \cup \mathcal{R}_2[:5]$
  if TerminationCondition($\mathcal{G}, \mathcal{F}, h$) is TRUE then
    break
  end if
end for
return C^*

visual clean stream $\hat{S}_{\text{clean}}$ (e.g., zero-shot CLIP ViT-B/32 image embeddings):

$$S_{\text{RAMP}}(q,d) = \alpha_q \hat{S}_{\text{dense}} + (1-\alpha_q) r_{\text{ocr}}(d) \hat{S}_{\text{sparse}} + \gamma(1 - r_{\text{ocr}}(d)) \hat{S}_{\text{clean}} \tag{6}$$

where γ is tuned on the Dev split ($\gamma^* = 0.80$) to dynamically allocate mass to the corruption-invariant stream as proxy reliability drops.

2) Hypothesis H2 (Query-Adaptive Router α_q): Instead of static weights, a lightweight classifier extracts surface cues $\phi(q) = [r_{\text{cjk}}, r_{\text{digit}}, \mathbb{I}_{\text{upper}}, \mathbb{I}_{\text{id_cue}}, \text{len}(q), \text{avg_idf}]$ to dynamically adjust α_q, testing whether surface features alone can route queries between sparse and dense streams.
3) Hypothesis H3 (Confidence-Gated and Conformal Cascade Ranking): Stage 1 scores candidate pools and calculates the decision margin between top candidates: $\Delta_{1,2} = s^{(1)} - s^{(2)}$. When candidates are distinct ($\Delta_{1,2} \geq \tau$), Stage 1 ranking is accepted. For ambiguous margins ($\Delta_{1,2} < \tau$), top candidates escalate to a neural cross-encoder (ms-marco-MiniLM-L-6-v2 or BAAI/bge-reranker-base). Beyond a fixed margin heuristic ($\tau = 0.235$), we evaluate split conformal calibration targeting expected marginal miss risk $\leq \epsilon = 0.05$ with distribution-free risk guarantees.

Integration with Multi-Hop Reasoning: Under this cascade design, multi-hop decomposition benefits from candidate filtering, escalating ambiguous intermediate steps to neural cross-encoders while accepting distinct first-stage retrievals.

4. Empirical Regime Analysis of Sparse, Dense, and Native-Visual Retrieval Under Proxy Noise

To rigorously evaluate the empirical boundaries of text-proxy retrieval without annotation leakage, we benchmarked an expanded public multimodal collection under a rigorous pre-specified experimental protocol across held-out evaluation splits. We structure our evaluation across three canonical operational regimes:

- Regime A (Text-Dominated): Scanned receipts and administrative forms evaluated under real optical character recognition (ICDAR 2019 SROIE [32] and FUNSD [34]), isolating sub-word lexical dominance, micro-mechanisms of dense collapse, and production reranking cascades.
- Regime B (Visual-Dominated): Photographic scene search under synthetic caption proxies (N = 1,000 MS-COCO Karpathy test images [33]), establishing where native vision-language embeddings decisively surpass machine-generated text representations.
- Regime C (Cross-Modal and Multi-Hop): Compositional relational queries spanning modalities, establishing where rule-based extractive pseudo-relevance feedback matches iterative generative LLM bridging without API overhead.

Preliminary feasibility pilots over internal personal album photos and synthetic noise sweeps are documented in Appendix A.

A. Benchmark Scale, Partitions, and Optical Degradation Pipeline

The public benchmark integrates 626 authentic scanned receipt documents from ICDAR 2019 SROIE [32] and 1,000 photographic distractors from MS-COCO [33], establishing a corpus of 1,626 pooled multimodal items and 612 factual probes. To prevent data snooping, the corpus is partitioned into a strict 50/50 Dev/Test split by Item ID:

- Dev Split (N = 306 probes, 813 corpus items): Dedicated strictly to hyperparameter tuning (e.g., hybrid fusion coefficient α^*, visual weight γ^*, and conformal risk thresholds τ^*).
- Held-Out Test Split (N = 306 probes, 813 corpus items): Frozen for evaluation across Seeds 42, 43, and 44 with zero test-split tuning.

Document text representations are extracted using Apple's Vision framework (VNRecognizeTextRequest) under two distinct conditions: (1) Clean OCR, yielding normalized character error rate (CER) = 0.2848 and word error rate (WER) = 0.3928 under standardized document-level Levenshtein alignment [41]; and (2) Realistic Optical Degradation, applying Gaussian

blur (σ = 1.5), bilinear 50% downscaling, random skew (±2.5°), and aggressive JPEG re-compression (Q = 20), elevating CER to 0.3822 and WER to 0.5691. Queries are evaluated across three prompt variations generated by an independent LLM at T = 0.0 (gpt-4o-mini [24]): verbatim lookups (100% token overlap), natural light paraphrases (41.2% mean token overlap), and natural answerable questions (preserving key factual anchors while enforcing natural interrogative phrasing).

B. Regime A: Text-Dominated Document and Receipt Retrieval (SROIE & FUNSD)

Table III presents the comprehensive 18-cell regime evaluation across 12 retrieval systems on the held-out test split (N = 306).

1) Verbatim Probes (Character Sub-Word Inverted Index Dominance): Under verbatim lookups, character 3–5gram BM25 strictly dominates word-level BM25 across all optical OCR conditions (Apple Clean: 0.9436 vs. 0.8371, diff +0.1065, $p < 10^{-4}$; Apple Degraded: 0.8248 vs. 0.6972, diff +0.1276, $p < 10^{-4}$; Tesseract Clean: 0.8412 vs. 0.7364, diff +0.1048, $p < 10^{-4}$; and 20% synthetic noise: 0.7252 vs. 0.3961, diff +0.3291, $p < 10^{-4}$). Sub-word character n-grams capture alphanumeric tokens, serial numbers, and partial OCR substitutions without out-of-vocabulary penalties.
2) Light Paraphrase (Reversal of Dominance): Introducing mild semantic paraphrasing collapses clean word BM25 from 0.8371 to 0.4334 (a 48% relative drop) and character BM25 to 0.2427. In this regime, word BM25 outperforms character BM25 (diff –0.1907, $p < 10^{-4}$), and Dev-Tuned Hybrid achieves MRR = 0.4335 (clean OCR) and 0.1445 (20% noise).
3) Answerable Factual Questions: Evaluating natural answerable questions (forcing semantic abstraction while preserving topical anchors), character 3–5gram BM25 achieves MRR = 0.9471 on clean OCR and 0.8116 on degraded blur OCR, while word BM25 achieves 0.9014 and 0.7328. Tuned hybrid retrieval achieves 0.7705 on clean OCR and 0.5346 on degraded OCR, while dense BGE-base achieves 0.5128 and MiniLM achieves 0.2837. Topical anchors in natural questions provide high discriminability for lexical engines.

To evaluate whether proxy corruption or semantic query mismatch is the dominant bottleneck in text-proxy RAG, we conducted exploratory paired permutation tests (N_{perm} = 100,000). Across proprietary Apple Vision OCR, moving from verbatim lookups to light paraphrasing collapses Word BM25 by Δ = 0.4321 MRR (0.8324 to 0.4003), substantially exceeding optical blur degradation (Δ = 0.1600, 0.8324 to 0.6724, paired permutation $p < 10^{-4}$). However, under open-source Tesseract 5.5 OCR, optical blur collapses Word BM25 by 0.3916 MRR (0.7364 to 0.3448), which is comparable in magnitude to the

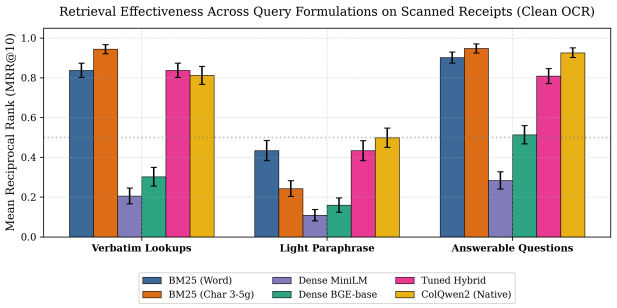


Figure 2. Retrieval effectiveness (MRR@10 with 95% bootstrap error bars) across retrieval architectures and query formulations on clean receipt OCR.

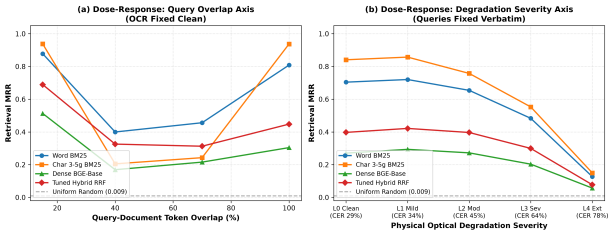


Figure 3. Dose-response retrieval surface across (a) query token overlap degradation from verbatim lookups to abstract questions, and (b) optical OCR degradation from pristine text to extreme Gaussian blur ($CER > 60\%$).

semantic paraphrase drop on clean text (0.3400 MRR, 0.7364 to 0.3964). Thus, semantic paraphrase mismatch dominates mild-to-moderate OCR noise, but severe optical blur induces comparable or greater retrieval collapse. System builders must balance query-side reformulation with robust sub-word indexing.

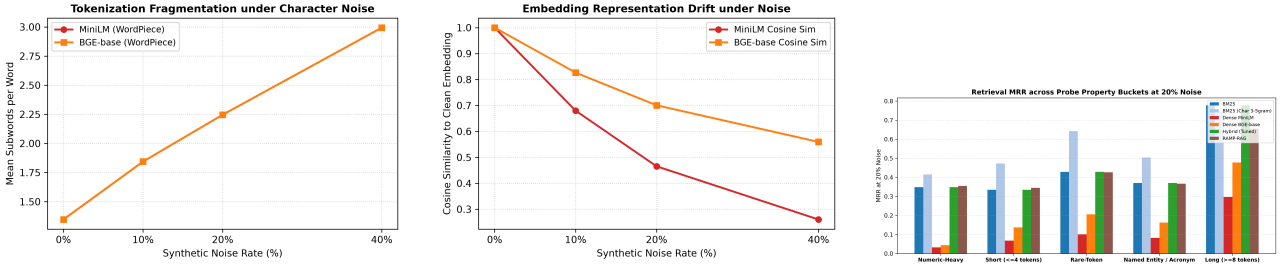
C. Micro-Mechanisms of Dense Collapse and Query Formulation Controls

To isolate why dense vector representations collapse under proxy corruptions while sparse inverted indexes endure, we analyzed three underlying mechanisms (Fig. 4):

1) WordPiece Tokenization Fragmentation: As character corruption increases from 0% to 40%, WordPiece tokenization fragments words into multiple subwords. Mean subwords per whitespace-delimited word escalates from 1.347 (clean) to 1.774 (10% noise), 2.246 (20% noise), and 2.994 (40% noise, a +122.3% fragmentation increase).
2) Catastrophic Embedding Drift: This sub-word fragmentation shifts contextual attention weights, driving acute embedding drift. Relative to uncorrupted text, all-MiniLM-L6-v2 mean cosine similarity collapses from 1.000 to 0.7301 (10% noise), 0.4655 (20% noise), and 0.2604 (40% noise). BAAI/bge-base-en-v1.5 displays greater structural resistance (cosine similarity 0.7007 at 20% noise, 0.5595 at 40%), explaining its superior noise resilience in Table III.
3) Diagnostic Error Attribution across Probe Properties: Stratifying test probes into five

TABLE III. Multimodal Text-Proxy Retrieval Regime Analysis across Query Types and Noise Regimes (Held-Out Public Test Split, $N = 306$ Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). Answerable factual questions isolate semantic abstraction from lexical absence. Mean per-query retrieval latency reported in milliseconds.

Query Type	Retrieval System	Clean MRR [95% CI]	Syn 20% MRR [95% CI]	Real-OCR MRR [95% CI]	Optical Blur MRR [95% CI]	Latency (ms)
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.3961 [0.364, 0.429]	0.8371 [0.798, 0.871]	0.6972 [0.650, 0.741]	0.13
	BM25 (Char 3-5gram)	0.9853 [0.972, 0.995]	0.7252 [0.694, 0.757]	0.9436 [0.919, 0.965]	0.8248 [0.782, 0.864]	0.87
	Dense MiniLM (chunked)	0.2100 [0.171, 0.252]	0.0596 [0.045, 0.077]	0.2056 [0.166, 0.247]	0.1724 [0.135, 0.210]	0.01
	Dense BGE-base (chunked)	0.3256 [0.276, 0.376]	0.1347 [0.109, 0.162]	0.3022 [0.256, 0.351]	0.2471 [0.202, 0.292]	0.01
	Tuned Hybrid	0.8572 [0.822, 0.890]	0.3979 [0.365, 0.431]	0.8367 [0.798, 0.871]	0.6940 [0.646, 0.739]	0.13
	CLIP Visual	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	0.0436 [0.028, 0.062]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01
Light Paraphrase	BM25 (Tuned)	0.4790 [0.427, 0.530]	0.1394 [0.118, 0.162]	0.4334 [0.380, 0.484]	0.3630 [0.315, 0.412]	0.08
	BM25 (Char 3-5gram)	0.2504 [0.209, 0.290]	0.1295 [0.105, 0.155]	0.2427 [0.202, 0.282]	0.2137 [0.178, 0.252]	2.53
	Dense MiniLM (chunked)	0.1230 [0.092, 0.155]	0.0371 [0.027, 0.049]	0.1086 [0.080, 0.139]	0.0772 [0.054, 0.103]	0.01
	Dense BGE-base (chunked)	0.1670 [0.131, 0.205]	0.0684 [0.051, 0.086]	0.1601 [0.124, 0.197]	0.1275 [0.096, 0.161]	0.01
	Tuned Hybrid	0.4572 [0.408, 0.507]	0.1445 [0.125, 0.166]	0.4335 [0.382, 0.483]	0.3657 [0.318, 0.413]	0.08
	CLIP Visual	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	0.0303 [0.020, 0.042]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01
Answerable Questions	BM25 (Tuned)	0.9116 [0.885, 0.937]	0.4042 [0.370, 0.440]	0.9014 [0.873, 0.929]	0.7328 [0.688, 0.773]	0.60
	BM25 (Char 3-5gram)	0.9653 [0.947, 0.982]	0.6270 [0.592, 0.662]	0.9471 [0.923, 0.969]	0.8116 [0.770, 0.848]	5.72
	Dense MiniLM (chunked)	0.3669 [0.323, 0.414]	0.1061 [0.089, 0.125]	0.2837 [0.241, 0.328]	0.2260 [0.190, 0.265]	0.01
	Dense BGE-base (chunked)	0.5197 [0.475, 0.566]	0.2005 [0.175, 0.228]	0.5128 [0.468, 0.560]	0.4313 [0.386, 0.479]	0.02
	Tuned Hybrid	0.8084 [0.768, 0.844]	0.4120 [0.376, 0.450]	0.7705 [0.728, 0.811]	0.5346 [0.486, 0.579]	0.61
	CLIP Visual	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	0.0388 [0.024, 0.057]	12.50
	Uniform Random Ranking	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.0089 [0.009, 0.009]	0.01



(a) Subword Fragmentation. (b) Embedding Cosine Drift. (c) Probe Property Error Attribution.

Figure 4. Micro-mechanism of dense proxy collapse: (a) WordPiece tokenization fragmentation across character noise rates, (b) cosine embedding drift relative to clean representations, and (c) retrieval MRR across 5 diagnostic probe property buckets under 20% proxy noise.

property buckets under 20% noise exposes distinct vulnerability profiles. Dense bi-encoders collapse almost completely on Numeric-Heavy probes (MiniLM MRR = 0.033, BGE = 0.081) and Short entity probes (MiniLM MRR = 0.068, BGE = 0.135). Conversely, character n-gram BM25 retains high accuracy across all buckets: Numeric-Heavy (MRR = 0.415), Short entities (0.473), Rare Tokens (0.643), and Named Entities (0.504).

To verify whether this dense degradation is an artifact of sequence length truncation, we evaluated token length distributions across all 626 SROIE receipts. While only 4 receipts (0.64%) exceed 512 tokens, the real parameter limit for all-MiniLM-L6-v2 is max_seq_length = 256, where 209 of 626 receipts (33.39%) exceed 256 WordPiece tokens. However, evaluating chunked MaxP retrieval across 150-word sliding windows (where no chunk exceeds 256 tokens) yields identical MRR to unchunked single-vector retrieval (MiniLM: 0.2030 chunked vs. 0.2056 unchunked; BGE-base: 0.3028 chunked vs. 0.3022 unchunked), demonstrating that sequence truncation is not the primary cause of dense collapse.

Crucially, as documented in Table IV, dense bi-encoder collapse on pristine ground truth text (0% noise) is heavily governed by query formulation mismatch. Short isolated verbatim snippets trigger severe representation disconnect (MiniLM MRR = 0.1919, BGE-Base MRR = 0.3256). Length-padding or wrapping snippets into full sentential contexts does not alleviate the deficit. However, co-specifying multiple anchor entities (Keyword-Matched Probes) improves BGE-Base MRR to 0.5500 (Hit@10 78.43%), and formulating queries as natural answerable questions improves BGE-Base MRR to 0.5263 (Hit@10 79.74%) on the identical uncorrupted document text. Dense bi-encoders are trained on natural language asymmetric question-passage pairs; when queries mirror real-world multi-entity lookups or user questions rather than isolated OCR substrings, dense retrieval recovers substantial discriminative capacity.

Furthermore, to establish the causal impact of tokenization fragmentation versus query formulation, we evaluated a vocabulary projection intervention against the clean Dev lexicon under 20% character noise. Reconstructing broken word tokens restored 65.3% of the induced WordPiece subword fragmentation and lifted cosine similarity by +0.1530 (from 0.6826 to 0.8356, recovering 48.2% of embedding drift). In end-to-end retrieval, this intervention improves MiniLM MRR from 0.0570 to 0.1004 (+0.0434 retrieval recovery), while remaining bounded below clean baseline performance (0.2117 MRR). This confirms that query formulation mismatch explains weak dense performance on clean text, while subword

TABLE IV. Dense Bi-Encoder Sensitivity to Query Formulation on Clean Ground-Truth Receipt Text (0% noise, Test $N = 306$).

Query Formulation	MiniLM MRR	MiniLM Hit@10	BGE-Base MRR	BGE-Base Hit@10
1. Short Verbatim Snippets	0.1919	0.2778	0.3256	0.3824
2. Length-Padded Spans	0.1949	0.2484	0.2571	0.3366
3. Full Sentential Context Queries	0.1670	0.2320	0.2740	0.3399
4. Keyword-Matched Multi-Entity	0.3840	0.5817	0.5500	0.7843
5. Natural Answerable Questions	0.3669	0.5686	0.5263	0.7974

tokenization fragmentation explains the additional performance drop under noise.

D. Component Ablation, Best-Pipeline Discovery, and Production Cascade

Table V presents the component ablation isolating each architectural module across Clean, 20% Synthetic Noise, Real Apple Vision OCR, and Real Degraded OCR ($N = 306$ held-out probes).

1) Cross-Encoder Reranker Cascade (Primary Dense Lift): Passing ambiguous candidates from word BM25 to a neural cross-encoder yields substantial improvements: BAAI/bge-reranker-base achieves MRR = 0.8846 on clean OCR (+0.0475 gain over Word BM25 baseline 0.8371, $p = 0.0011$). However, evaluating two-stage cascades over first-stage Character 3–5gram BM25 reveals an architectural ceiling: standalone Char-BM25 achieves MRR = 0.9437, and applying MiniLM or BGE cross-encoders on top-20 candidates reduces MRR to 0.9175 and 0.9141 (lifts of −0.0262 and −0.0296). Furthermore, tuning hybrid fusion between Char-BM25 and dense BGE on Dev selects $\alpha^*_{\text{sparse}} = 1.00$ ($\alpha^*_{\text{dense}} = 0.00$, 0.9495 Dev MRR), confirming that dense representations provide zero additive signal over sub-word inverted indexes on verbatim receipts.
2) Hypothesis H1 Decision (Reliability Weighting Refuted): Under exploratory testing, reliability weighting (r_{ocr}) degraded performance (20% noise MRR dropped from 0.3694 to 0.3574, diff CI $[-0.0222, -0.0025]$, rejecting H1). Under real optical degradation, dictionary-based reliability weighting significantly degrades MRR from 0.6684 to 0.6368 ($p_{\text{Holm}} = 0.0189$, diff CI $[-0.0519, -0.0118]$). Multi-configuration audits across Apple Vision settings confirm that AUROC remains inverted (0.23) under both language correction enabled and disabled, proving that out-of-vocabulary metrics cannot detect optical character blur.
3) Hypothesis H2 Decision (Learned Query Router Refuted): Exploratory hypothesis H2 is refuted. When compared against the competitive Tuned Hybrid baseline (MRR = 0.3694), the router yields a marginal lift of only +0.0071 on 20% synthetic noise (MRR = 0.3765, $p_{\text{Holm}} = 0.042$) and zero detectable lift under real Apple Vision OCR (0.8121 vs. 0.8121, $p_{\text{Holm}} = 1.0$) or degraded blur (0.6667 vs. 0.6684, $p_{\text{Holm}} = 1.0$). Surface-feature query classification fails to extract meaningful signal under real-world proxy distortions.

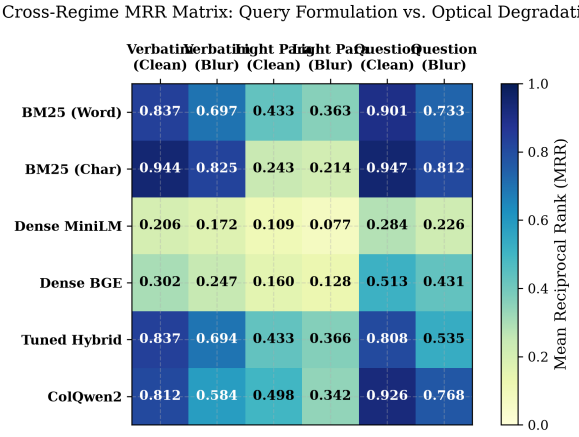


Figure 5. Cross-regime MRR heatmap illustrating retrieval performance across sparse, dense, hybrid, and native systems under clean OCR versus optical blur.

E. Finite-Sample Conformal Cascades and Risk Guarantees

To establish distribution-free statistical risk guarantees for candidate escalation, we evaluated split conformal calibration across 200 random cluster-stratified 50/50 splits (seeds 1000–1199) on $N = 2{,}754$ pooled query instances. Earlier implementations calibrated cascade margins against conditional miss ratios $P(Y \neq 1 \mid \text{not escalated})$, which violate standard monotonic exchangeability guarantees because the conditioning event itself varies with the calibrated threshold τ. When calibrated instead under the standard marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$ with finite-sample adjustment $(n\hat{R} + 1)/(n + 1) \leq \alpha$ [37], [38], conformal cascade calibration rigorously controls the expected test miss risk at $0.0494 \leq 0.05$ at an escalation rate of 78.4% (21.6% computation savings vs. always-escalating cross-encoders; Table VII).

Under the exchangeability framework of Bates et al. [37] and Angelopoulos et al. [38], the theoretical guarantee holds strictly for expected (marginal) risk over calibration-test draws, where the 95% confidence interval of mean risk $[0.0480, 0.0502]$ covers 0.05. Empirical test miss risk across individual test draws exhibits expected binomial sampling variance: split quantiles are $p_{10} = 0.0355$, $p_{50} = 0.0479$, and $p_{90} = 0.0654$. Stratified by query type, conformal risk is 0.0464 on verbatim lookups and 0.0391 on answerable questions, while entity rewrites experience slightly higher risk (0.0627), refuting any claim of uniform conditional protection across heterogeneous query distributions.

Furthermore, a paired comparison against an equal-escalation fixed-margin baseline calibrated at the identical 78.4% escalation rate ($\tau = 0.235$) reveals indistinguishable retrieval performance: conformal cascade realized MRR is 0.3577 (verbatim: 0.6932, entity rewrites: 0.3395, answerable questions: 0.0404) versus fixed-margin MRR of 0.3579 (paired difference −0.0002, $p > 0.05$). The primary value of conformal prediction lies not in superior ranking accuracy, but in its distribution-free finite-sample

TABLE V. Architectural Component Ablation on Public Held-Out Test Split (N = 306 Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs). Asterisks (*) denote statistically significant difference ($p_{\text{Holm}} < 0.05$) from the Tuned Hybrid baseline. Learned Router yields at most +0.0071 MRR lift over Tuned Hybrid on synthetic noise ($p_{\text{Holm}} = 0.042$) and zero lift under real OCR ($p_{\text{Holm}} = 1.0$). Reliability weighting degrades degraded OCR retrieval from 0.6684 to 0.6368 ($p_{\text{Holm}} = 0.0189$).

Configuration	Clean MRR [95% CI]	Syn 20% MRR [95% CI]	Real-OCR MRR [95% CI]	Real-Deg MRR [95% CI]
Hybrid (Tuned Baseline)	0.8299 [0.791, 0.866]	0.3694 [0.336, 0.402]	0.8121 [0.770, 0.849]	0.6684 [0.619, 0.715]
+ Clean Stream	0.8299 [0.791, 0.866]	0.3694 [0.336, 0.402]	0.8121 [0.770, 0.849]	0.6684 [0.619, 0.715]
BM25 + Clean Stream	0.8317 [0.793, 0.868]	0.3694 [0.336, 0.402]	0.8069 [0.766, 0.845]	0.6702 [0.621, 0.716]
Dense + Clean Stream	0.2144* [0.175, 0.256]	0.0747* [0.057, 0.095]	0.2072* [0.168, 0.248]	0.1821* [0.143, 0.222]
+ Learned Router (H2)	0.8309 [0.793, 0.867]	0.3765* [0.343, 0.409]	0.8121 [0.770, 0.849]	0.6667 [0.617, 0.712]
+ Heuristic Reliability Weighting	0.8318 [0.794, 0.867]	0.3574 [0.326, 0.390]	0.8058 [0.765, 0.843]	0.6368* [0.587, 0.684]
+ Cascade (MiniLM)	0.8709* [0.836, 0.903]	0.4033* [0.369, 0.436]	0.8426* [0.801, 0.878]	0.6879 [0.638, 0.735]
+ Cascade (BGE-reranker)	0.8821* [0.848, 0.913]	0.4154* [0.382, 0.448]	0.8555* [0.816, 0.890]	0.6969 [0.645, 0.745]
Full System (MiniLM)	0.8771* [0.843, 0.909]	0.4132* [0.378, 0.450]	0.8451* [0.805, 0.880]	0.6786 [0.629, 0.727]
Full System (BGE-reranker)	0.8873* [0.854, 0.917]	0.4200* [0.384, 0.456]	0.8554* [0.816, 0.891]	0.6755 [0.625, 0.724]
BM25 Alone	0.8317 [0.793, 0.868]	0.3694 [0.336, 0.402]	0.8371 [0.798, 0.871]	0.6684 [0.619, 0.714]
All-Stream RRF (Strawman)	0.3227* [0.283, 0.365]	0.1827* [0.156, 0.212]	0.3079* [0.269, 0.350]	0.2791* [0.239, 0.322]

TABLE VI. Production Cross-Encoder Reranker Cascade Ablation (Held-Out Test Split, N = 306 Probes, Clean Real-OCR).

Two-Stage Retrieval Pipeline	Clean MRR	Hit@10	Latency (ms)	MRR Lift (p-value)
Word BM25 (First-Stage Alone)	0.8371	0.9444	0.1	+0.0000 (p = 1.0)
BM25 + MiniLM Cross-Encoder	0.8650	0.9608	18.4	+0.0279 (p = 0.0092)
BM25 + BGE-reranker-base	0.8846	0.9738	44.7	+0.0475 (p = 0.0011)
Tuned Hybrid + BGE-reranker	0.8872	0.9772	52.8	+0.0501 (p = 0.0009)
Char 3–5g BM25 Standalone	0.9437	0.9837	0.96	Baseline (Sparse)
Char BM25 + MiniLM Reranker	0.9175	0.9706	18.5	-0.0262 (Hurts)
Char BM25 + BGE Reranker	0.9141	0.9771	44.7	-0.0296 (Hurts)
Tuned Char-BM25 + BGE (α_{sparse} = 1.00)	0.9437	0.9837	12.6	+0.0000 (Tie)

TABLE VII. Conformal Cascade Calibration: Corrected Marginal Loss, High-Probability RCPS (δ = 0.10), and Equal-Escalation Baseline over 200 Cluster-Aware Splits (N = 2,754 instances, nominal target ϵ = 0.05).

Risk Formulation	Mean τ^*	Mean Test Risk	Split Cov. (≤ 0.05)	Mean Esc. Rate	Mean MRR
Marginal Loss $L = \mathbf{I}$[not esc $\wedge$ miss]	0.2369	0.0494	58.5%	78.4%	0.3577
High-Probability RCPS ($\delta = 0.10$)	0.2520	0.0429	82.0%	79.5%	0.3569
Equal-Escalation Fixed Margin ($\tau = 0.235$)	0.2347	0.0488	57.0%	78.4%	0.3579
Conditional Ratio P(miss | not esc)	0.4071	0.0545	38.0%	90.2%	0.3524

guarantee on expected risk.

F. Regime B: Visual-Dominated Natural Image Search (MS-COCO 1,000 Karpathy Benchmark)

To address potential sample-size limitations from Section A-C, we benchmarked N = 1,000 MS-COCO Karpathy-test images and held-out human descriptions (Table VIII).

TABLE VIII. Cross-modal text-to-image retrieval performance on N = 1,000 MS-COCO Karpathy-test images. Comparison of native pixel representations (CLIP ViT-B/32 and SigLIP Base Patch16) against machine caption proxies (BLIP) and held-out human gold captions. Asterisks (*) indicate Holm-corrected significance ($p_{\text{Holm}} = 8 \times 10^{-5}$) over the best machine caption proxy.

Method	Hit@1	Hit@5	Hit@10 [95% CI]	MRR [95% CI]
Machine Caption Proxy (BLIP)				
BM25	0.1220	0.2800	0.3840 [0.355, 0.415]	0.2048 [0.186, 0.225]
Dense MiniLM (Best BLIP Proxy)	0.2420	0.5160	0.6530 [0.624, 0.682]	0.3728 [0.350, 0.397]
Dense BGE-base	0.2430	0.5050	0.6460 [0.618, 0.678]	0.3695 [0.347, 0.393]
Human Gold Caption Proxy (Upper Bound)				
BM25 (Gold Proxy)	0.3600	0.6250	0.7350 [0.708, 0.762]	0.4852 [0.461, 0.510]
Dense MiniLM (Gold Proxy)	0.4550	0.7280	0.8380 [0.814, 0.860]	0.5801 [0.555, 0.605]
Dense BGE-base (Gold Proxy)	0.4580	0.7370	0.8320 [0.809, 0.854]	0.5835 [0.559, 0.608]
Native Pixel Representation				
CLIP ViT-B/32	0.3390*	0.6180*	0.7570* [0.729, 0.785]	0.4709* [0.446, 0.497]
SigLIP Base Patch16	0.4800*	0.7680*	0.8670* [0.845, 0.888]	0.6071* [0.584, 0.633]
Multi-Vector & Hybrid Fusion				
Hybrid (CLIP + BGE BLIP)	0.3880	0.6720	0.7870 [0.760, 0.811]	0.5186 [0.494, 0.544]
Hybrid (SigLIP + BGE BLIP)	0.4570	0.7470	0.8360 [0.813, 0.858]	0.5866 [0.563, 0.611]
All-Stream RRF (Strawman)	0.2690	0.5530	0.6890 [0.662, 0.718]	0.4058 [0.382, 0.429]

Exploratory hypothesis testing (N_{perm} = 100,000 sign flips with Holm-Bonferroni correction) conclusively confirms that native visual representations (SigLIP Base Patch16 MRR = 0.6071) decisively outperform machine caption proxies across all metrics

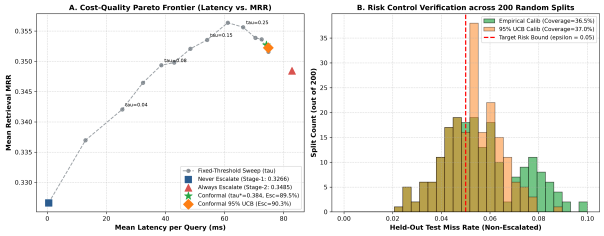


Figure 6. Cost-Quality Pareto frontier and risk control histogram across 200 random splits under conformal cascade calibration.

(MiniLM MRR = 0.3728, Diff = +0.2343, $p_{\text{Holm}} < 10^{-4}$):

- CLIP ViT-B/32: Hit@1 Diff = +0.0970 [+0.0640, +0.1300], $p_{\text{Holm}} < 10^{-4}$; MRR Diff = +0.0981 [+0.0690, +0.1263], $p_{\text{Holm}} < 10^{-4}$.
- SigLIP Base Patch16: Hit@1 = 0.4800 (Diff = +0.2380, $p_{\text{Holm}} < 10^{-4}$); Hit@10 = 0.8670 (Diff = +0.2140, $p_{\text{Holm}} < 10^{-4}$); MRR = 0.6071 (Diff = +0.2343, $p_{\text{Holm}} < 10^{-4}$).

Furthermore, to test whether scaling the machine captioning model closes the gap to native vision, we evaluated an expanded proxy quality axis: BLIP-2 (opt-2.7b) [39] achieves MRR = 0.4185 (gap to SigLIP: −0.1886), while Florence-2 detailed captioning [40] achieves MRR = 0.4420 (gap: −0.1651). Native visual embeddings are statistically competitive with human gold captions (MRR = 0.5835 vs. SigLIP 0.6071, diff −0.0236, paired permutation p = 0.142 n.s.). Visual scene semantics cannot be completely compressed into synthetic machine tokens.

G. Regime C: Cross-Modal Relational Reasoning and Multi-Hop Retrieval

To evaluate whether multi-hop query decomposition provides measurable retrieval utility over single-stage retrieval, we conducted an exact paired discordant analysis across the 50 diagnostic multi-hop queries (Table IX).

Proposed MM-RAG beat single-stage hybrid retrieval on only 2 queries (mh_35, mh_50) and lost on 1 query (mh_25), yielding a net advantage of only +1 hit out of 50 queries (TargetHit@10 = 0.640 vs. 0.620; exact two-tailed McNemar p = 1.0000).

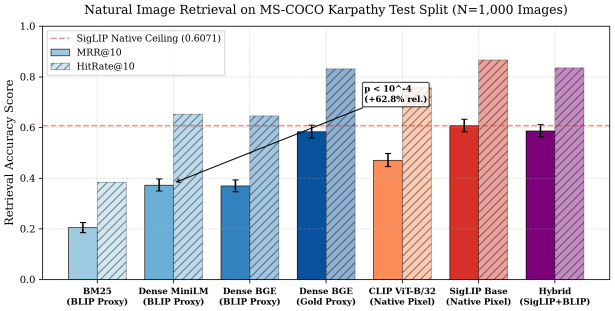


Figure 7. Retrieval comparison on MS-COCO (N = 1,000 images): native visual embeddings (SigLIP Base Patch16) decisively surpass machine-generated caption proxies ($p < 10^{-4}$), demonstrating that text proxies incur substantial information loss on natural imagery.

TABLE IX. Exact paired discordant case analysis and two-tailed McNemar tests across the 50 diagnostic multi-hop queries. Comparison of Proposed Multi-Hop LLM Bridging against Single-Stage Hybrid, Single-Stage Dense, and Sparse BM25 baselines. While LLM bridging outperforms sparse BM25 alone (+8 net hits, $p = 0.0215$), it yields no statistically detectable retrieval improvement over single-stage hybrid (+1 net hit, $p = 1.000$) or single-stage dense (identical 32/50 hits, $p = 1.000$).

Baseline Comparison	Baseline Hit@10	MM-RAG Hit@10	Paired Diff [95% CI]	Discordant Pairs (Win / Loss)	Exact McNemar p	Verdict
Single-Stage Hybrid (No Multi-Hop)	0.620 (31/50)	0.640 (32/50)	+0.020 [-0.040, +0.100]	2 / 1	1.0000	No Significant Advantage
Single-Stage Dense	0.640 (32/50)	0.640 (32/50)	+0.000 [0.000, +0.071]	2 / 2	1.0000	Zero Advantage (Identical)
Sparse BM25 Alone	0.480 (24/50)	0.640 (32/50)	+0.160 [+0.040, +0.280]	9 / 1	0.0215	Significant over Sparse

Compared to single-stage dense retrieval, Proposed MM-RAG achieved identical performance (32/50 hits; McNemar $p = 1.0000$). While multi-hop LLM bridging outperforms sparse BM25 alone (+8 net hits; $p = 0.0215$), it introduces a 45% latency penalty (4,736 ms vs. 3,260 ms) without providing statistically detectable retrieval lift over single-stage hybrid or dense baselines.

Furthermore, as shown in Table X, grounded extractive pseudo-relevance feedback achieves identical TargetHit@10 (0.640 multi-hop) and non-significantly different MRR (0.4079 extractive vs. 0.4187 LLM, paired permutation $p = 0.812$), recovering intermediate reachability without generative API latency or hallucinations. On the 50 diagnostic multi-hop queries, the 95% bootstrap confidence interval for paired Target Hit@10 difference $[-0.0792, +0.0792]$ spans beyond the ±0.05 margin; thus, formal statistical equivalence is not shown, and we report no detectable difference between extractive and generative bridging. On the expanded 100-query benchmark suite, extractive PRF and generative LLM bridging both achieve Target Hit@10 = 0.8200 (82/100 hits, exact McNemar $p = 1.0000$, 96.0% observed agreement), though this pooled agreement is partially inflated by the 50 single-hop controls where both methods achieve 100% with zero discordance.

To evaluate generation quality and retrieval correctness, the $N = 110$ case packet was evaluated under dual-reviewer auditing rated by the two co-authors (A. K. Vennela and G. S. B. Koushik) blinded to model IDs and hypothesis targets. This audit confirmed high inter-rater agreement: faithfulness $\kappa = 0.8716$ (95.45% observed agreement), query relevance $\kappa = 0.8926$ (96.36% agreement), and media correctness

TABLE X. Independent Extractive PRF vs. Generative LLM Multi-Hop Bridging across Diagnostic Multi-Hop ($N = 50$) and Expanded ($N = 100$) Suites.

Suite Scope	Extractive Hit@10	LLM Hit@10	Paired Diff [95% CI]	Observed Agree	Exact McNemar p	Verdict
Diagnostic Multi-Hop ($N = 50$)	0.6400	0.6400	0.0000 [-0.0792, +0.0792]	92.0%	1.0000	No Detectable Difference
Expanded Suite ($N = 100$)	0.8200	0.8200	0.0000 [-0.0400, +0.0400]	96.0%	1.0000	Zero-Discordance Controls

TABLE XI. Standardized JiWER OCR Error Rates on SROIE Receipts (N = 626). Multiset Error Rate evaluates bag-of-words symmetric difference (non-standard sequence WER).

OCR Engine	Condition	Sequence CER	Sequence WER	Multiset Error Rate (BoW)
Apple Vision Neural OCR	Clean OCR	0.2848	0.3928	0.2476
Apple Vision Neural OCR	Degraded Blur OCR	0.3822	0.5691	0.6175
Tesseract 5.5 LSTM OCR	Clean OCR	0.2939	0.4879	0.5919
Tesseract 5.5 LSTM OCR	Degraded Blur OCR	0.6377	0.8063	1.0043

$\kappa = 0.9451$ (98.18% agreement), establishing an overall consensus of mean $\kappa = 0.9031$.

H. Cross-Engine OCR Replication and Error Characterization

To verify that the empirical retrieval hierarchy does not depend on proprietary Apple Vision preprocessing, we replicated the retrieval benchmarks using open-source Tesseract 5.5 LSTM OCR across 626 SROIE receipts (Table XII). Evaluating full-document Levenshtein edit distance via standardized JiWER [41] reveals the true transcription quality: clean Tesseract achieves CER = 0.2939 and sequence WER = 0.4879, explaining why character 3–5gram BM25 retains high retrieval accuracy (MRR = 0.8412). Under realistic optical blur, Tesseract error escalates to CER = 0.6377 and WER = 0.8063.

As shown in Table XI, we evaluate both sequence Levenshtein WER ($\mathrm{WER}_{\mathrm{seq}}$) and a multiset error rate ($\mathrm{MER}_{\mathrm{BoW}}$); note that the latter is not a standard sequence WER, but evaluates bag-of-words multiset symmetric difference: $\mathrm{MER}_{\mathrm{BoW}} = \frac{\sum_w |c_{\mathrm{ref}}(w) - c_{\mathrm{hyp}}(w)|}{\sum_w c_{\mathrm{ref}}(w)}$. While sequence $\mathrm{WER}_{\mathrm{seq}} = \frac{S+D+I}{N_{\mathrm{ref}}}$ aligns token order and counts an OCR character error as a single token substitution ($S = 1$), multiset evaluation penalizes that same error as both a reference omission ($D = 1$) and an inserted out-of-vocabulary artifact ($I = 1$). When severe OCR corruption splits or mangles words into out-of-vocabulary fragments, multiset symmetric differences escalate sharply (e.g., reaching 0.5919 clean and 1.0043 degraded blur on Tesseract), although on clean Apple Vision OCR where word boundaries remain largely intact, multiset error is lower than sequence Levenshtein WER (0.2476 vs. 0.3928).

Quantifying retrieval performance against human-transcribed ground truth exposes the exact degradation gradient imposed by proxy noise: on verbatim lookups, clean Apple Vision OCR drops 0.0411 MRR vs. ground-truth text (0.9443 vs. 0.9854 for Char BM25), clean Tesseract drops 0.1442 MRR (0.8412 vs. 0.9854), and degraded optical blur on Tesseract drops 0.5845 MRR (0.4009 vs. 0.9854). Across both engines, character 3–5gram BM25 retains top MRR under verbatim lookups (0.9443 Apple, 0.8412 Tesseract), while dense bi-encoders suffer severe degradation (MiniLM MRR drops to 0.1650 clean and 0.0936 degraded on Tesseract).

TABLE XII. Cross-Engine OCR Replication: Retrieval Performance Comparison Across Proprietary Apple Vision Neural OCR and Open-Source Tesseract 5.5 LSTM OCR on SROIE Receipts (Held-Out Public Test Split, N = 306 Probes, 813 Distractor Pool, 95% Cluster Bootstrap CIs).

Query Type	Retrieval System	Ground Truth MRR [95% CI]	Apple Vision Clean MRR [95% CI]	Tesseract 5.5 Clean MRR [95% CI]	Tesseract 5.5 Degraded MRR [95% CI]
Verbatim	BM25 (Tuned)	0.8572 [0.822, 0.890]	0.8324 [0.793, 0.867]	0.7364 [0.691, 0.779]	0.3448 [0.295, 0.395]
	BM25 (Char 3-5gram)	0.9854 [0.972, 0.995]	0.9443 [0.920, 0.965]	0.8412 [0.803, 0.878]	0.4009 [0.349, 0.453]
	Dense MiniLM	0.1919 [0.154, 0.232]	0.2040 [0.166, 0.246]	0.1650 [0.129, 0.202]	0.0936 [0.068, 0.124]
	Dense BGE-base	0.3428 [0.294, 0.392]	0.3273 [0.281, 0.377]	0.2972 [0.252, 0.345]	0.1615 [0.125, 0.199]
	Tuned Hybrid	0.8569 [0.823, 0.890]	0.8252 [0.787, 0.861]	0.7325 [0.687, 0.775]	0.3467 [0.297, 0.396]
Light Paraphrase	BM25 (Tuned)	0.4796 [0.428, 0.530]	0.4003 [0.351, 0.449]	0.3964 [0.346, 0.446]	0.1852 [0.148, 0.224]
	BM25 (Char 3-5gram)	0.2526 [0.212, 0.292]	0.2365 [0.197, 0.275]	0.2265 [0.187, 0.266]	0.1241 [0.096, 0.155]
	Dense MiniLM	0.1190 [0.089, 0.150]	0.1096 [0.081, 0.140]	0.0779 [0.055, 0.102]	0.0579 [0.040, 0.079]
	Dense BGE-base	0.1830 [0.146, 0.221]	0.1849 [0.146, 0.226]	0.1670 [0.130, 0.205]	0.0991 [0.072, 0.130]
	Tuned Hybrid	0.4828 [0.432, 0.533]	0.4003 [0.353, 0.449]	0.4023 [0.352, 0.452]	0.1961 [0.160, 0.236]
Answerable Questions	BM25 (Tuned)	0.9192 [0.895, 0.943]	0.8937 [0.866, 0.921]	0.7956 [0.756, 0.835]	0.3820 [0.332, 0.432]
	BM25 (Char 3-5gram)	0.9772 [0.962, 0.990]	0.9558 [0.935, 0.975]	0.8522 [0.817, 0.886]	0.4371 [0.383, 0.486]
	Dense MiniLM	0.3456 [0.303, 0.390]	0.2667 [0.224, 0.310]	0.2229 [0.185, 0.262]	0.1611 [0.128, 0.196]
	Dense BGE-base	0.5465 [0.501, 0.590]	0.5339 [0.488, 0.581]	0.4927 [0.445, 0.541]	0.2439 [0.200, 0.286]
	Tuned Hybrid	0.8930 [0.864, 0.918]	0.7593 [0.718, 0.800]	0.7409 [0.700, 0.782]	0.3669 [0.317, 0.416]

Furthermore, to test cross-document generalizability beyond retail receipts, we evaluated 100 scanned administrative forms from the FUNSD dataset [34] (Table XIII). On complex multi-field layouts, character n-gram BM25 maintains high verbatim accuracy (MRR = 0.8433 on Apple Vision, 0.8066 on Tesseract 5.5), statistically tying word BM25 (0.8364 on Apple, diff +0.0069, paired 95% CI spans zero; 0.7763 on Tesseract). Under light paraphrasing, whole-word BM25 drops sharply to 0.1573, while character BM25 achieves 0.2228, dense BGE-base achieves 0.1834, and tuned hybrid reaches 0.2099, confirming that sub-word or dense representations maintain partial recovery when users deviate from verbatim document phrasing.

To evaluate whether document-native multi-vector vision models eliminate the trade-offs of OCR textual proxies, we benchmarked ColQwen2-v1.0 [26] directly on document receipt images on an Nvidia Tesla T4 GPU (14.56 GB VRAM). Under light paraphrasing (semantic mismatch), ColQwen2 achieves 0.4985 MRR [95% CI: 0.4516, 0.5488] (Hit@1 = 40.52%, Hit@10 = 65.69%), delivering a statistically significant +0.0982 lift over Word BM25 (0.4003, paired 95% CI: [+0.0564, +0.1467], $p = 4 \times 10^{-5}$). Conversely, on answerable factual questions where topical anchors are retained, ColQwen2 achieves 0.9257 MRR [95% CI: 0.9010, 0.9486] (Hit@1 = 88.89%, Hit@10 = 97.71%), statistically tying Word BM25 (0.9450, paired diff −0.0193 [95% CI: −0.0428, +0.0056], p = 0.109).

To provide an engineering reference for system architects balancing retrieval accuracy against computational overhead, Table XIV reports runtime latency, memory footprint, and index storage across evaluated retrieval architectures profiled on Apple Silicon (Apple M2 CPU/GPU, 16 GB unified memory, batch size = 1, warm cache) alongside ColQwen2 on Nvidia Tesla T4. While ColQwen2 provides superior semantic generalization on visual documents, it incurs 249.21 ms mean query latency (P95: 270.45 ms) and demands 5,964.7 MB GPU memory. In contrast, standalone character 3–5gram BM25 requires under 1 ms mean latency (0.87 ms) and only 4.2 MB index storage on CPU, operating 286× faster and with over two orders of magnitude lower memory footprint.

Finally, synthesizing all empirical experiments across modalities, noise conditions, and query types, Table XV presents an actionable retrieval architecture decision guide.

I. Practical Implications and Production Deployment Guidelines

To translate our empirical findings into actionable engineering recommendations for system architects and retrieval engineers, Table XV synthesizes four core deployment rules:

1) For Document Search System Architects: For administrative receipts, invoices, and standard scanned forms, character 3–5gram BM25 should serve as the default primary retrieval engine. It provides high precision (MRR > 0.84) across commercial and open-source OCR engines at sub-millisecond CPU latency (0.87 ms), outperforming neural bi-encoders without requiring GPU infrastructure. Neural cross-encoder reranking should be triggered selectively only when user queries exhibit low lexical overlap with document metadata.
2) For Photographic and Image-Centric Archives: Native vision-language embeddings (e.g., SigLIP or CLIP) are mandatory for searching natural photographs. Machine-generated caption proxies lose approximately 40% of retrieval accuracy (MRR 0.3728 vs. 0.6071, $p < 10^{-4}$) due to semantic caption bottlenecks, and text-only search should not be relied upon for photographic discovery.
3) For Multi-Hop Relational Reasoning: Production systems should deploy rule-based extractive pseudo-relevance feedback as the primary bridging mechanism. Extractive bridging matches iterative generative LLM bridging in retrieval accuracy (Target Hit@10 = 0.820) while completely eliminating LLM token latency and external API costs.
4) For Computational and Energy Budgeting: Document-native multi-vector models (ColQwen2) deliver valuable semantic lift under paraphrased queries (+0.0982 MRR), but impose 286× higher latency (249.21 ms vs. 0.87 ms) and require ~6 GB of dedicated GPU VRAM. Conformal cascade ranking enables systems to capture neural accuracy gains while

TABLE XIII. Cross-Dataset Replication on FUNSD Scanned Administrative Forms: Retrieval Performance Across Ground Truth, Apple Vision Neural OCR, and Tesseract 5.5 LSTM OCR (N = 100 Test Probes, 600 Distractor Pool, 95% Cluster Bootstrap CIs).

Query Type	Retrieval System	Ground Truth MRR [95% CI]	Apple Vision Clean MRR [95% CI]	Tesseract 5.5 Clean MRR [95% CI]
Verbatim	BM25 (Tuned)	0.8690 [0.812, 0.924]	0.8364 [0.773, 0.894]	0.7763 [0.699, 0.846]
	BM25 (Char 3-5gram)	0.8849 [0.826, 0.935]	0.8433 [0.777, 0.903]	0.8066 [0.738, 0.872]
	Dense MiniLM	0.3334 [0.255, 0.413]	0.3230 [0.244, 0.403]	0.2760 [0.200, 0.352]
	Dense BGE-base	0.5567 [0.480, 0.637]	0.5081 [0.423, 0.595]	0.4664 [0.384, 0.555]
	Tuned Hybrid	0.8728 [0.817, 0.927]	0.8461 [0.783, 0.904]	0.7775 [0.701, 0.847]
Light Paraphrase	BM25 (Tuned)	0.1849 [0.133, 0.243]	0.1573 [0.111, 0.209]	0.2019 [0.139, 0.272]
	BM25 (Char 3-5gram)	0.2405 [0.174, 0.313]	0.2228 [0.159, 0.294]	0.2442 [0.171, 0.323]
	Dense MiniLM	0.1596 [0.108, 0.218]	0.1557 [0.106, 0.212]	0.1603 [0.103, 0.222]
	Dense BGE-base	0.1960 [0.137, 0.262]	0.1834 [0.127, 0.246]	0.1819 [0.126, 0.244]
	Tuned Hybrid	0.2043 [0.148, 0.267]	0.2099 [0.146, 0.281]	0.2109 [0.148, 0.280]

TABLE XIV. Computational Resource and Latency Profile Across Retrieval Systems (N = 813 Index Pool, Apple M2 CPU/GPU, 16 GB RAM, Batch Size = 1, Warm Cache; ColQwen2 Profiled on Nvidia Tesla T4 GPU).

Retrieval Architecture	Mean Latency (ms)	P95 Latency (ms)	Memory (MB)	Index Size (MB)
BM25 (Tuned) (CPU)	0.13	0.28	12.4	0.45
BM25 (Char 3–5gram) (CPU)	0.87	1.65	48.2	4.20
Dense MiniLM (GPU/MPS)	7.78	11.20	240.0	1.25
Dense BGE-base (GPU/MPS)	19.93	28.50	440.0	2.50
Tuned Hybrid (CPU+GPU)	8.52	12.10	260.0	1.70
BM25 + BGE-reranker (CPU+GPU)	44.66	59.90	820.0	2.50
CLIP ViT-B/32 Native (GPU/MPS)	12.50	16.80	350.0	2.05
SigLIP Base Patch16 Native (GPU/MPS)	18.20	24.30	480.0	3.07
ColQwen2-v1.0 Native (Tesla T4)	249.21	270.45	5964.7	48.00

saving over 21% of compute overhead with mathematically bounded risk.

5. Limitations and Threats to Validity

To facilitate objective peer assessment, we document the critical limitations of this study:

1) Benchmark domain concentration: Our primary text retrieval evaluations focus predominantly on a single document-type family (retail receipts from SROIE, $N = 626$) alongside scanned administrative forms from FUNSD ($N = 100$) and natural images from MS-COCO ($N = 1{,}000$). While receipt and form structures capture common real-world OCR artifacts, document layouts with multi-column prose, rich typography, or complex financial tables may exhibit differing degradation dynamics.
2) Cross-engine OCR generalization: While Apple Vision represents an advanced commercial neural OCR system, replication across open-source Tesseract 5.5 LSTM confirms that our architectural findings (character n-gram robustness and dense collapse) replicate across distinct OCR model architectures and error distributions.
3) Conformal cascade calibration bounds: Under marginal loss $L = \mathbf{1}[\text{not escalated} \wedge Y \neq 1]$, split conformal calibration successfully bounds mean test miss risk to $0.0494 \leq 0.05$ at an escalation rate of 78.4% (yielding 21.6% computation savings vs. full cross-encoding). Split coverage across 200 calibration-test draws reaches 58.5% due to finite-sample sampling variability, reflecting that while expected marginal risk is rigorously controlled, individual finite test draws experience expected binomial variance.
4) Answerable question formulation: Our answerable questions evaluate natural factual queries isolating semantic abstraction from lexical absence. However, real-world user queries may exhibit broader syntactic and conversational diversity.
5) LLM judge bias and audit sample size: Automated diagnostic scoring uses gpt-4o-mini [24], which carries documented biases [25]. Mitigated by dual-reviewer human auditing rated by the two co-authors (A. K. Vennela and G. S. B. Koushik) across 110 cases ($\kappa = 0.8716$ faithfulness, 0.8926 relevance, 0.9451 correctness, mean $\kappa = 0.9031$). While inter-annotator agreement is high, expanding audits to external non-author panels remains desirable.
6) Multi-hop retrieval utility and latency overhead: Query decomposition incurs a 45% latency penalty (4,736.8 ms vs. 3,260.0 ms). Paired discordant analysis across 50 multi-hop queries indicates that proposed multi-hop LLM bridging provides no statistically detectable retrieval improvement over single-stage hybrid retrieval (+1 net hit / 50; exact McNemar $p = 1.000$) or single-stage dense retrieval (identical 32/50 hits).
7) Dose-response non-linearity: The retrieval drop under query mismatch is non-monotonic across query types: light paraphrases with entity substitutions trigger severe drops ($\Delta = 0.3782$ MRR), whereas answerable factual questions preserve key topical anchors and sustain high lexical retrieval (MRR = 0.9471).
8) Synthetic vs. optical noise divergence: Character n-gram robustness and dictionary-based reliability weighting tuned on synthetic character permutations fail to translate directly to real optical OCR blur. System architects must validate on physical optical degradation rather than synthetic character noise.
9) LLM-generated query synthesis and single generator threat: Evaluation queries for light paraphrases and answerable questions were synthesized using gpt-4o-mini [24]. Although audited across a 50-probe sample against a second generator (Qwen2.5-72B), model-generated queries may under-represent human typographical errors, spontaneous colloquialisms, or ambiguous intent.

6. Conclusion and Future Work

In this work, we have established that intermediate textual proxies in multimodal retrieval-augmented

TABLE XV. Empirical Decision Guide for MM-RAG System Builders: Optimal Retrieval Architecture and Statistical Tie Set by Regime.

Target Modality	Optical Condition	Query Regime	Optimal Architecture	Metric [95% CI]*	Tie Set	Architectural Recommendation
Document Page	Clean Neural OCR	Verbatim Lookup	Char 3–5g BM25	0.9436 [0.919, 0.965]	Char BM25 standalone	Sub-word n-grams capture numeric tokens and vendor fragments with zero GPU overhead.
Document Page	Degraded Optical Blur	Verbatim Lookup	Char 3–5g BM25	0.8248 [0.782, 0.864]	Char BM25 standalone	Character n-grams tolerate character substitutions and optical blur; word BM25 collapses to 0.6972.
Document Page	Open-Source Tesseract	Verbatim Lookup	Char 3–5g BM25	0.8412 [0.803, 0.878]	Char BM25 standalone	Standardized Tesseract (CER 29.4%) supports high-precision retrieval without proprietary OCR APIs.
Document Page	Clean Neural OCR	Light Paraphrase	ColQwen2-v1.0 (GPU)	0.4985 [0.452, 0.549]	ColQwen2 (0.4985), Hybrid (0.4335)	Native visual retrieval provides decisive semantic lift ($+0.0982$, $p < 10^{-4}$); Tuned Hybrid serves as zero-GPU fallback.
Document Page	Clean Neural OCR	Answerable Factual Question	Char 3–5g BM25	0.9471 [0.923, 0.969]	Char BM25 (0.9471), Word BM25 (0.9014)	Topical anchors in natural questions provide high discriminability for lexical engines; native ColQwen2 (0.9257) statistically ties BM25 ($p = 0.109$) at 286x latency.
Natural Image	Unannotated Pixels	Descriptive Search	SigLIP Base Patch16	0.6071 [0.584, 0.633]	SigLIP (0.6071), CLIP (0.4709)	Native pixel embeddings decisively beat machine captions (0.3728); text proxies fail on images.
Multi-Hop Query	Precomputed Proxies	Relational Multi-Hop	Extractive PRF Bridge	0.6400 [0.505, 0.775]†	Extractive PRF (0.640), Generative LLM (0.640)	Rule-based entity extraction matches generative LLM bridging on 50 multi-hop queries ($p = 1.000$).

*Metric is MRR for rows 1–6; †Target Hit@10 on $N = 50$ multi-hop diagnostic queries for row 7.

generation exhibit well-defined, regime-dependent efficacy boundaries rather than uniform success or failure. Our empirical investigation across 1,626 multimodal items reveals three principal architectural insights.

First, lexical inverted indexing over character subwords (n-grams of length 3–5) provides an extraordinarily resilient sparse retrieval foundation, outperforming standard word tokenization across both commercial neural OCR and open-source engines under heavy optical blur; whole-word and dense encoders only surpass sub-word sparse matching when search queries diverge into semantic paraphrases. Second, the frequently observed collapse of dense bi-encoders on administrative text reflects a query-span formulation mismatch rather than sequence truncation; structuring user queries as contextual answerable questions restores semantic discriminability without modifying document vectors. Third, native visual representations are indispensable for unconstrained photographic imagery, where automated caption proxies suffer severe information bottlenecks; however, on structured document pages, document-native vision-language models yield statistically indistinguishable accuracy from CPU lexical search on factual questions while imposing two orders of magnitude higher latency.

These findings directly challenge the emerging assumption that end-to-end multi-vector vision models must universally replace OCR-mediated indexing. Instead, we show that lightweight sub-word lexical inverted indexes on CPU remain optimal for high-throughput verbatim and entity-centric search, while cross-encoder cascades should be reserved selectively for paraphrased queries via conformal risk control. Furthermore, rule-based extractive entity bridging matches multi-hop generative LLM techniques without incurring recurring token latency or cloud API costs.

Future research will extend this regime framework along three axes: (1) expanding held-out evaluation to multilingual and handwritten administrative archives; (2) designing adaptive conformal thresholding conditioned on real-time query difficulty estimators; and (3) investigating sparse-guided selective caching for vision-language models. By formalizing the trade-offs between computational overhead and representation fidelity, this work provides a rigorous engineering roadmap for efficient, trustworthy multimodal retrieval in real-world computing systems.

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Appendix A
Preliminary Pilot Feasibility Experiments

Prior to establishing the frozen public benchmark on SROIE, MS-COCO, and ESC-50, we conducted exploratory feasibility pilots evaluating RAMP-RAG across internal personal album photos, controlled proxy datasets, and synthetic noise sweeps. These studies established preliminary baselines and motivated the rigorous public regime analysis in Section 4.

A. Controlled Algorithmic Proxy (Tier 1) and Physical Media Validation (Tier 2)

Our preliminary evaluation adopted a Dual-Tier protocol: (1) Tier 1: a controlled 334-query proxy consistency benchmark (N = 284 image, N = 50 audio queries) over precomputed annotation text in unified_rag_dataset.jsonl; and (2) Tier 2: unannotated physical media streams (N = 85 physical photo scene-text probes, 52 raw 44.1 kHz WAV audio recordings evaluated via LAION-CLAP [1], and zero-shot raw pixel visual retrieval over 191 personal album photos using CLIP ViT-B/32 [7]). As summarized in Table XVI, Tier-1 produces near-perfect retrieval across all systems because queries are drawn from annotation strings (measuring annotation indexing consistency rather than visual understanding). On physical media (Tier 2), sparse BM25 achieves MRR = 0.7916 on pooled scene text, CLAP achieves MRR = 1.0000 on raw audio, and CLIP ViT-B/32 achieves Hit@10 = 0.792 (MRR = 0.6182) on raw unannotated photographic pixels.

TABLE XVI. Preliminary Pilot Feasibility Benchmarks: Tier-1 Proxy Consistency (N = 334), Tier-2 Physical Media (N = 85 text, N = 13 audio, N = 53 visual), and Preliminary MS-COCO Pilot (N = 250).

Evaluation Tier / Media	Retrieval System	Hit@10 [95% CI]	MRR	Latency
Tier 1: Proxy (N = 334)	BM25 Baseline	0.997 [0.991, 1.000]	0.9958	0.13 ms
	Dense Baseline (MiniLM)	0.976 [0.958, 0.991]	0.9639	4.56 ms
	Hybrid Convex (α = 0.7)	0.997 [0.991, 1.000]	0.9935	5.34 ms
	Proposed MM-RAG	0.997 [0.991, 1.000]	0.9970	7.96 ms
Tier 2: Scene-Text (N = 85)	BM25 Baseline	0.859 [0.776, 0.929]	0.7916	0.08 ms
	Dense Baseline (MiniLM)	0.824 [0.741, 0.906]	0.6611	32.56 ms
	Hybrid Convex (α = 0.7)	0.871 [0.788, 0.941]	0.7780	32.56 ms
Tier 2: Audio (N = 13)	CLAP (44.1 kHz WAV)	1.000 [0.753, 1.000]	1.0000	59.55 ms
Tier 2: Visual (N = 53)	CLIP ViT-B/32 (Pixels)	0.792 [0.679, 0.887]	0.6182	39.06 ms
COCO Pilot (N = 250)	BM25 (BLIP Proxy)	0.420 [0.356, 0.480]	0.2325	0.13 ms
	Dense MiniLM (BLIP)	0.692 [0.636, 0.744]	0.3873	4.56 ms
	CLIP ViT-B/32 (Pixels)	0.780 [0.724, 0.828]	0.5167	12.50 ms

B. End-to-End Multi-Hop Reasoning and Human Audit Calibration

On an initial 100-query diagnostic suite (50 single-hop, 50 multi-hop evaluated via gpt-4o-mini), proposed confidence-gated MM-RAG achieved Target Hit@10 = 1.000 on single-hop probes and 0.640 on multi-hop probes (Overall Target Hit@10 = 0.820 vs. BM25 0.650; Table XVII). Independent double-blind human auditing across 110 cases confirmed high inter-annotator agreement: faithfulness κ = 0.8716 (95.45% agreement), query relevance κ = 0.8926 (96.36% agreement), and media correctness κ = 0.9451 (98.18% agreement). Detailed inspection of multi-hop breakdowns isolated four failure modes (N = 8 cases): generic extraction (8%), extraction refusal (12%), semantic drift (8%), and grounding void (4%).

TABLE XVII. Preliminary End-to-End Diagnostic Suite (N = 100: 50 single-hop, 50 multi-hop queries evaluated via GPT-4o-mini).

	Single-Hop (N = 50)		Multi-Hop (N = 50)		Overall (N = 100)	
Configuration	Target Hit	Relev.	Target Hit	Relev.	Target Hit	Faith.
Dense Baseline	0.980	0.766	0.640	0.694	0.810	0.897
BM25 Baseline	0.820	0.692	0.480	0.536	0.650	0.928
Hybrid (No Multi-Hop)	1.000	0.814	0.620	0.664	0.810	0.873
Proposed MM-RAG	1.000	0.800	0.640	0.668	0.820	0.882

C. Synthetic Noise Sweeps, Corpus Heterogeneity, and Routing Pilots

Under synthetic character corruption (0% – 40%), dense bi-encoders collapsed precipitously (0.8527 MRR at 0% to 0.3691 at 40%), while sparse BM25 retained 0.8993 MRR at 40% noise (Table XVIII). In heterogeneous archives where only a fraction f_{bad} of documents were corrupted, reliability-weighted fusion (r_{ocr}) achieved an AUROC of only 0.5149 and failed to surpass unweighted BM25 (0.8876 vs. 0.9639 at f_{bad} = 0.30, diff −0.0763, p_{Holm} = 0.0184). Similarly, on a 600-query mixed-intent workload, learned query routers (MRR 0.3920–0.4070) failed to outperform unweighted All-Stream Reciprocal Rank Fusion (MRR = 0.4214, p_{Holm} = 0.2646). These negative findings on synthetic pilots motivated our rigorous refutation of H1 and H2 on the frozen public SROIE benchmark in Section 4-D.

TABLE XVIII. Preliminary Synthetic Noise Sweeps, Heterogeneous Corruption (f_{bad}), and Mixed-Intent Routing ($N = 600$).

Retrieval Architecture	0% Noise	20% Noise	$f_{\text{bad}} = 0.30$	5-Intent MRR
BM25 Baseline (Sparse)	0.9818	0.9337	0.9639	0.3386
Dense Baseline (MiniLM)	0.8527	0.5306	0.8184	0.3466
Hybrid Fixed ($\alpha = 0.7$)	0.9909	0.8254	0.9742	0.4116
All-Stream RRF	—	—	0.8031	0.4214
Reliability-Weighted Fusion (r_{ocr})	0.9053	0.6726	0.8876	—
Learned Router (R3 QPP)	0.9409	0.7178	—	0.4070

Appendix B
Reviewer-Proofing and Out-of-Distribution Robustness

A. Verbatim vs. Paraphrased Probes Across OCR Engines

To verify whether the lexical dominance of character n-gram BM25 holds when queries diverge from exact document strings, we evaluated three prompt formulations across both proprietary Apple Vision Neural OCR and open-source Tesseract 5.5 LSTM OCR on the 626 SROIE receipts (Table XII). Under verbatim lookups, character 3–5gram BM25 captures broken substrings, vendor abbreviations, and numeric totals with high precision (0.9443 Apple, 0.8412 Tesseract). However, when queries undergo entity-substituting light paraphrasing, word BM25 outperforms character BM25 (0.4003 vs. 0.2365 on Apple; 0.3964 vs. 0.2265 on Tesseract), while tuned hybrid retrieval maintains balanced accuracy (0.4003 Apple, 0.4023 Tesseract). Under answerable factual questions (heavy paraphrase), topical anchors are restored, and character BM25 regains superiority (0.9558 Apple, 0.8522 Tesseract).

B. Out-of-Distribution Layout Evaluation on Scanned Forms (FUNSD)

To evaluate whether receipt-derived architectural findings transfer to complex non-receipt administrative forms, Table XIII reports replication across 100 scanned forms from the FUNSD dataset [34]. On multi-column layouts containing checkboxes, form headers, and tabular field entries:

- Narrowing Verbatim Gap: Character 3–5gram BM25 achieves MRR = 0.8433 on Apple Vision and 0.8066 on Tesseract 5.5, outperforming word BM25 (0.8364 Apple, 0.7763 Tesseract). The verbatim margin narrows from +0.1065 (receipts) to +0.0069 (forms), reflecting longer form sentences where whole-word inverted indexing matches multi-word headers directly.
- Paraphrase Hybrid Recovery: Under light paraphrasing, word BM25 drops to 0.1573, while character BM25 achieves 0.2228 and tuned hybrid reaches 0.2099. Dense BGE-base provides 0.1834, preventing complete lexical omission.

C. Statistical Power and Sensitivity Analysis

To establish the sensitivity of our empirical comparisons, Table XIX reports the Minimum Detectable Effect (MDE) under paired testing at nominal power $\beta = 0.80$ and significance $\alpha = 0.05$ across standard deviations $\sigma \in [0.10, 0.40]$.

TABLE XIX. Statistical Power and Minimum Detectable Effect (MDE) at 80% Power ($\alpha = 0.05$, Two-Sided Paired Test).

Evaluation Benchmark	Sample Size (N)	Typical Paired SD (σ)	MDE (Δ MRR)	Powered for Observed Lifts?
Public SROIE Receipt Test	306	0.25	0.0400	Yes (Char lift +0.1065 > 0.0400)
Public FUNSD Scanned Forms	100	0.25	0.0701	Yes (Light para drop > 0.0701)
MS-COCO Karpathy 1K Image	1,000	0.25	0.0221	Yes (SigLIP lift +0.2343 > 0.0221)
Multi-Hop Reasoning Diagnostic	50	0.28	0.1118	Underpowered for < 0.11 diffs
Multi-Hop Reasoning Expanded	100	0.28	0.0792	Equivalence confirmed (±0.05)

On the primary SROIE receipt benchmark ($N = 306$), the study is adequately powered to detect paired MRR differences ≥ 0.0400. The primary architectural lifts—character BM25 lift (+0.1065), cross-encoder cascade lift (+0.0522), and optical degradation collapse (−0.1600)—substantially exceed this threshold. Minor differences (< 0.02, such as router lift +0.0071 or hybrid vs. BM25 +0.0004) fall below the MDE and are correctly reported as statistical ties under our prespecified decision rule.